



TRIBHUVAN UNIVERSITY
Institute of Science and Technology

A Project Proposal
On
“La Traino”
A Personalized Physical Training System

Submitted To
Department of Statistics and Computer Science
Patan Multiple Campus

*In partial fulfillment of the requirements for Bachelor Degree in Computer
science and Information Technology*

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1 Introduction

Physical fitness and healthy lifestyle have become increasingly important in modern society. However, access to professional fitness trainers and personalized workout routines is limited for many individuals due to cost, time, and location constraints. With the growth of web technologies, digital fitness platforms have emerged as an effective solution to bridge this gap.

This project proposes a multi-vendor fitness platform where multiple independent trainers provide personalized training services to multiple trainees through a centralized web application. La Traino is a web application that connects trainers and trainees through a digital platform. The system supports both self-guided trainees using preset workout routines and trainer-guided trainees who receive personalized workout and diet plans. The application is accessible across devices and offers flexibility, scalability, and ease of use.

2 Problem statement

Many individuals lack access to structured fitness guidance, while professional trainers face difficulties in managing multiple trainees and tracking their progress remotely. Existing fitness applications are either costly, platform-dependent, or lack personalization and real-time monitoring. There is a need for a lightweight, cross-platform system that supports both independent workouts and personalized trainer-led fitness management.

3 Objectives

The main objectives of La Traino are:

- To develop a web application that links trainers with their trainees digitally.
- To provide a standard preset workout routine for self-guided trainees.
- To provide features for trainers to assign personalized workout and diet plans.
- To track workout completion and progress of the trainee.
- To provide weekly progress visualization and trainer feedback.

4 Methodology

4.1 Requirement Identification

Requirements for the system will be gathered through study of existing fitness applications, online research, and user feedback from trainers and fitness enthusiasts.

4.1.1 Study of Existing System

Existing fitness systems often focus either on self-guided workouts or trainer-led programs but rarely integrate both effectively. Many applications lack real-time progress tracking and personalized feedback mechanisms, limiting user engagement.

4.1.2 Literature Review

Digital health and fitness platforms have gained widespread adoption due to increased smartphone usage and internet accessibility. Studies show that digital exercise platforms help reduce barriers related to cost, time, and location by enabling remote access to structured fitness guidance. These platforms support personalized workout delivery and long-term engagement through interactive features and self-monitoring tools ([Bennett et al., 2025](#)).

Personalization, progress tracking, and user engagement are key factors influencing fitness outcomes. Research indicates that personalized workout plans combined with regular progress monitoring significantly improve motivation, exercise adherence, and overall physical performance (Bennett et al., 2025). Additionally, mobile fitness applications that offer enjoyment, usefulness, and trainer interaction positively affect user wellbeing and sustained participation ([Kim et al., 2024](#)).

Technology-assisted fitness systems have also demonstrated measurable health benefits. Longitudinal studies show that consistent use of mobile fitness applications is associated with improved physical activity patterns and weight management (Furukawa et al., 2023). Furthermore, virtual trainer guidance and online coaching systems have been found to enhance exercise motivation and adherence, supporting the effectiveness of trainer-trainee interaction in digital fitness platforms ([Chen et al., 2024](#)).

4.1.3 Requirements Analysis

Requirements will be collected through interviews, Google searches, website visits and friendly suggestions.

4.1.3.1 Functional Requirement

- User registration and login (Trainer/Trainee)
- Preset workout routine access
- Trainer code generation and trainee linking
- Personalized workout and diet assignment

- Workout completion tracking
- Weekly progress charts and trainer feedback

4.1.3.2 Non-functional Requirement

- Availability - Usable across all devices and in all geographical locations.
- Reliability - The system will be reliable and data will be secure.
- User Interaction - Attractive and easy user interface.

4.2 Feasibility Study

4.2.1 Technical Feasibility

The system will be developed using following technologies:

- Frontend: React.js
- Backend: Node.js / Python
- Database: PostgreSQL

4.2.2 Operational Feasibility

As the application is web-based, it can run on desktops, tablets, and mobile devices through web browsers. No special hardware required.

4.2.3 Economic Feasibility

The system will be built using the tools that are freely available on the internet as well as royalty free images, so, this system is economically feasible.

4.3 High Level Design of System

4.3.1 Methodology of the proposed system

The project follows the Waterfall Model:

1. Requirement Analysis
2. System Design
3. Implementation
4. Testing
5. Documentation

4.3.2 System Flowchart

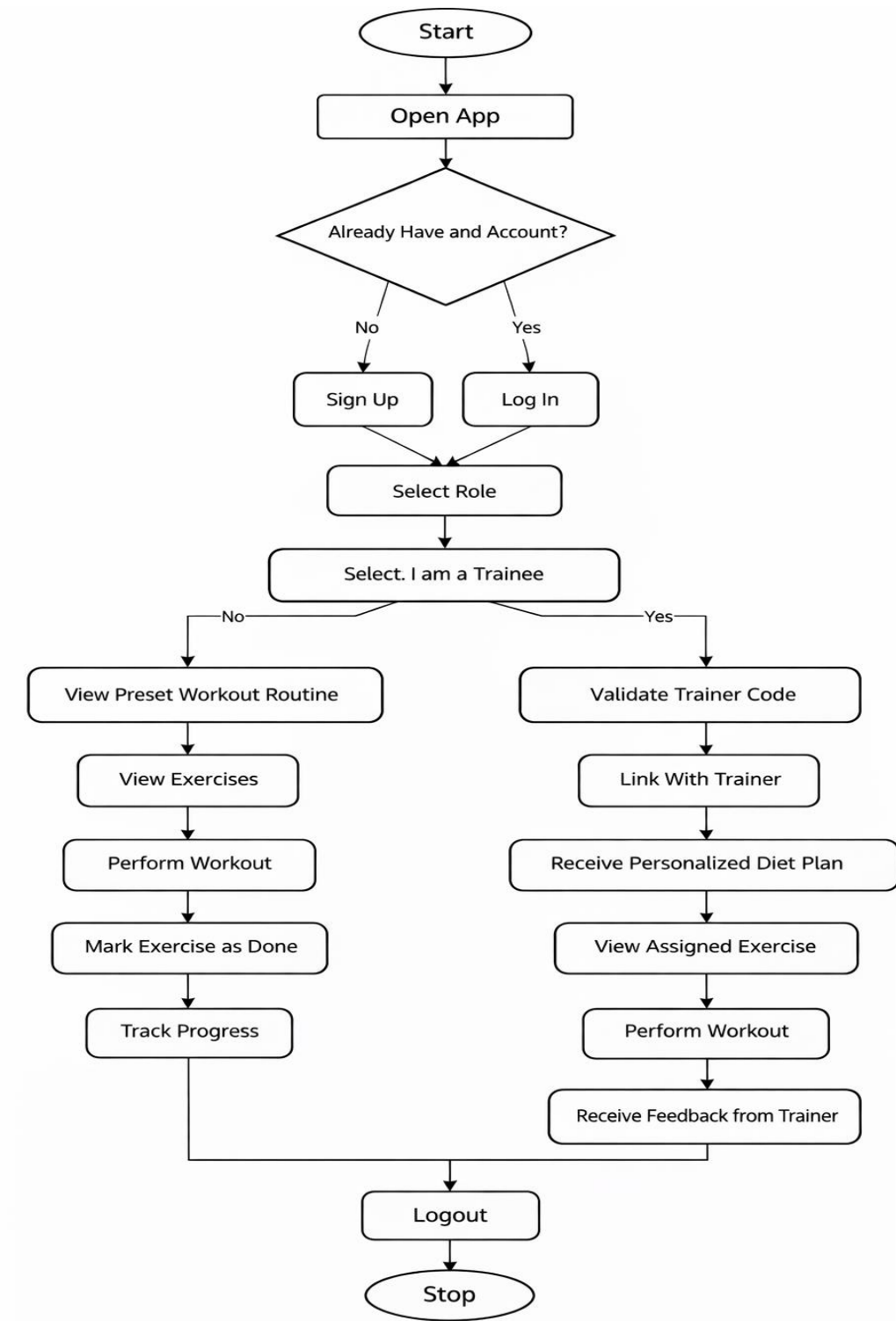


Figure 1: System Flowchart for User (Trainee)

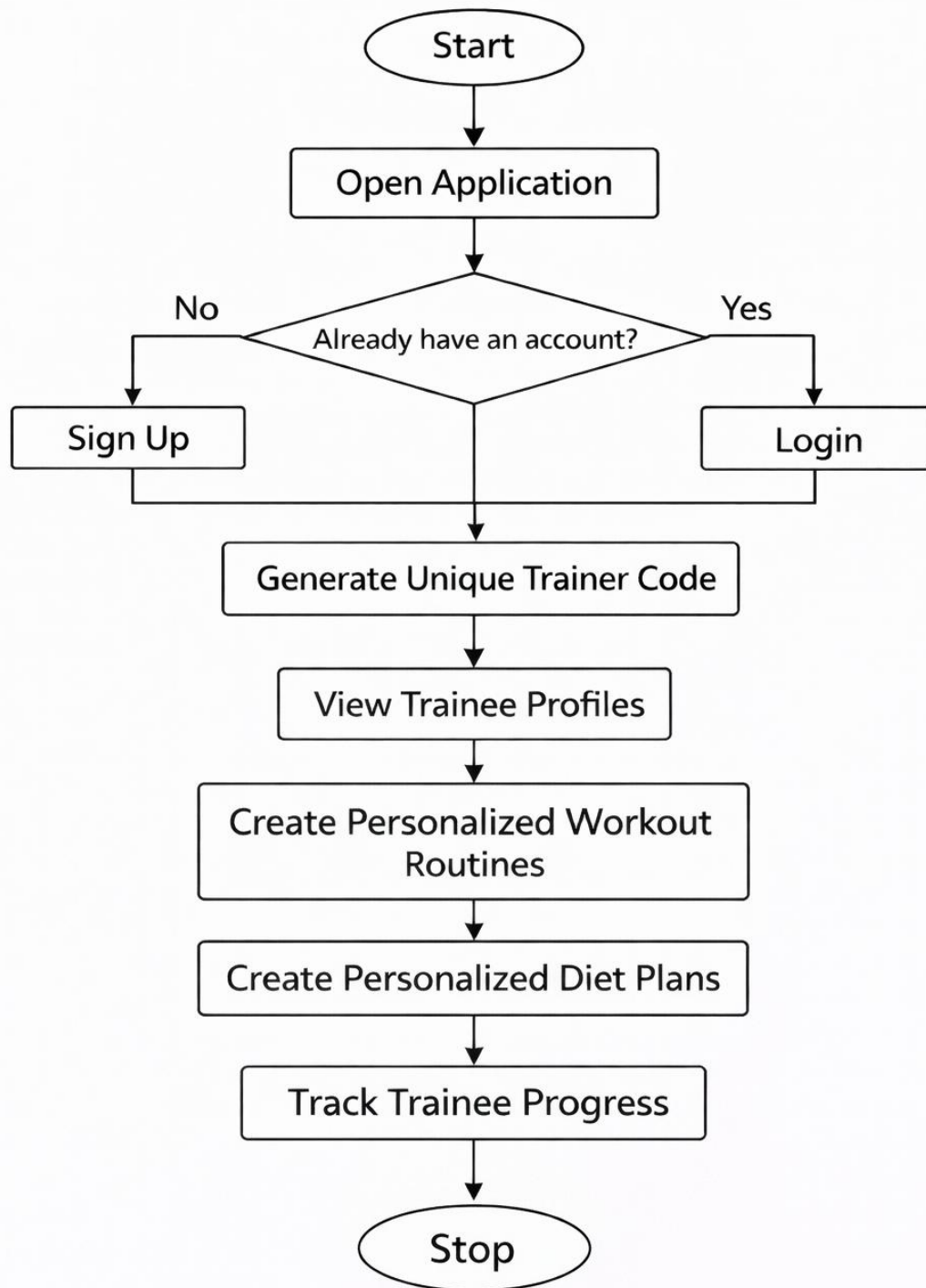


Figure 2: System Flowchart for User (Trainer)

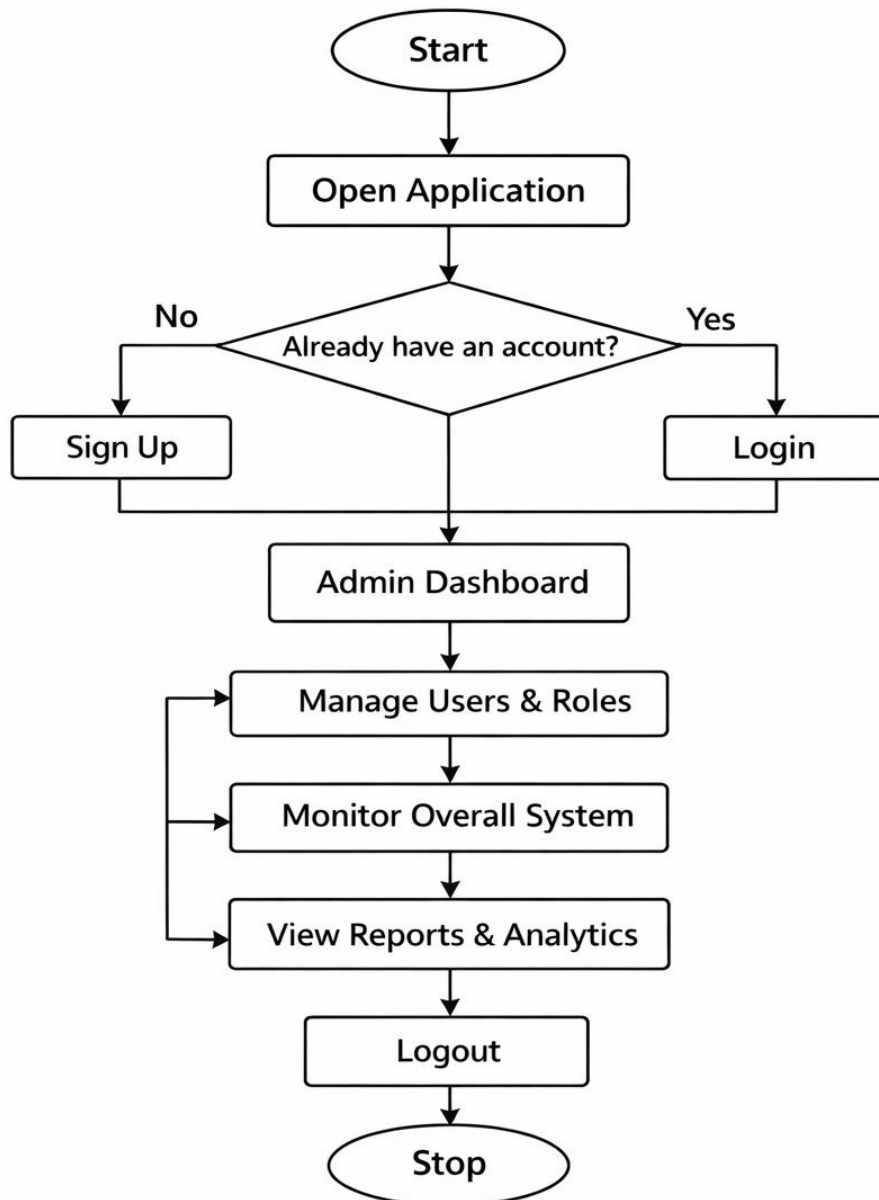


Figure 3: System Flowchart for Admin

4.3.3 Use Case Diagram

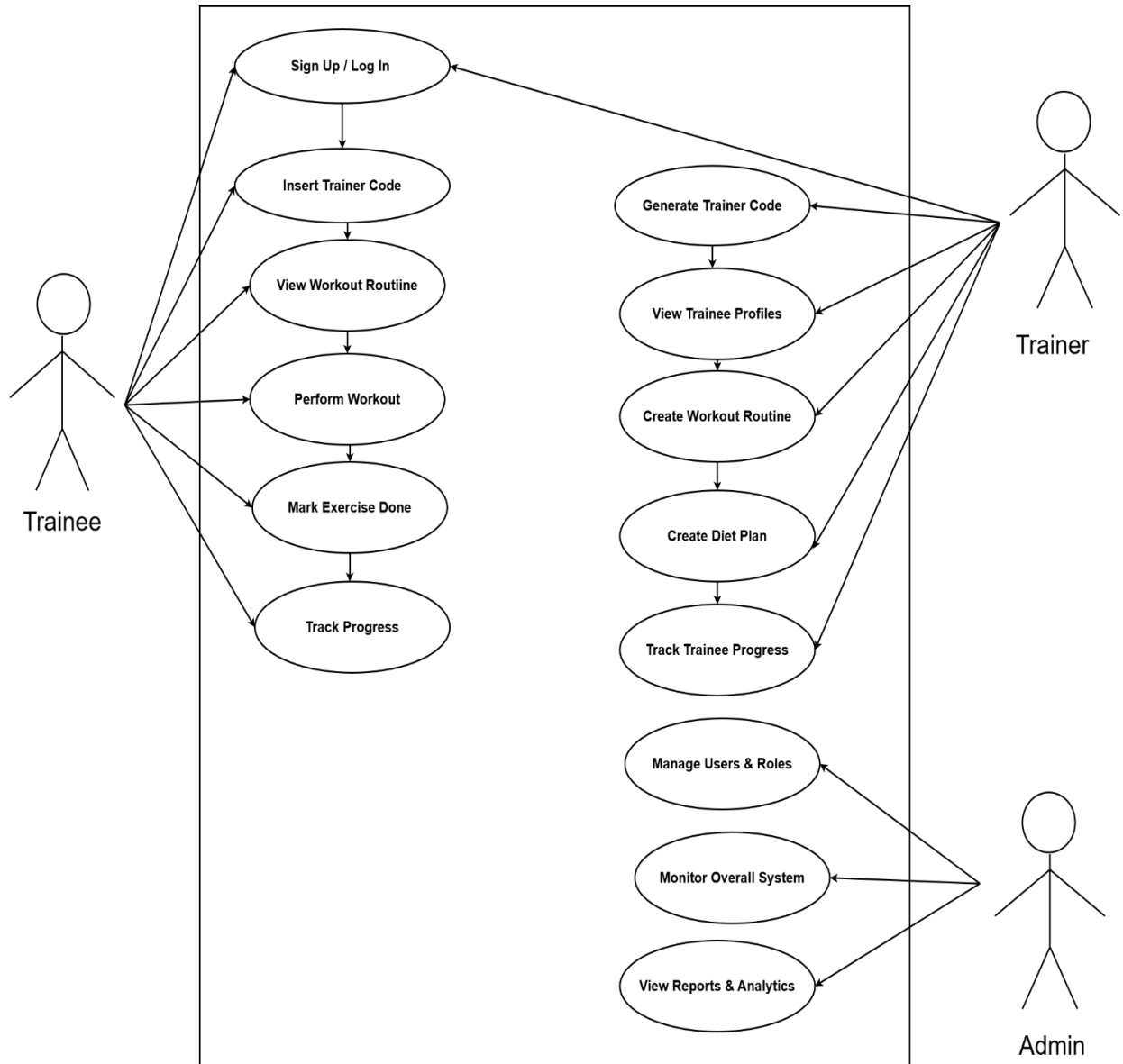


Figure 4: Use case diagram

4.3.4 Description of Algorithms

The proposed gym training application uses a Unique Trainer Code Generation and Validation algorithm to securely connect trainees with their respective trainers. The system generates a unique alphanumeric code for each trainer during registration. Trainees use this code to join and receive training services from the selected trainer. This approach ensures secure trainer-trainee linking, prevents unauthorized access, and enables efficient management of trainee assignments within the system.

Unique Trainer Code Generation and Validation

The Unique Trainer Code Generation and Validation algorithm generates a distinct identification code for each trainer and verifies the authenticity of the code when entered by trainees. Each trainer is assigned a randomly generated alphanumeric code consisting of letters and numbers. The system checks the generated code against existing trainer codes in the database to ensure uniqueness. When a trainee enters the trainer code, the system validates it by matching it with stored trainer records and establishes a trainer-trainee relationship.

Working of Unique Trainer Code Generation and Validation Algorithm

- Step 1: Start
- Step 2: Trainer registers and creates an account in the system
- Step 3: The system generates a random alphanumeric trainer code
- Step 4: The system checks the database to ensure that the generated code is unique
- Step 5: The trainer code is stored in the database along with trainer details
- Step 6: Trainee registers and enters the trainer code
- Step 7: The system verifies the trainer code by searching the database
- Step 8: If the code is valid, trainee is linked to the trainer and gains access to training plans
- Step 9: If the code is invalid, the system displays an error message and requests re-entry
- Step 10: Stop

Mathematically, the uniqueness of trainer codes depends on the number of possible combinations that can be generated. If the trainer code is generated using a combination of uppercase letters (A–Z) and numbers (0–9), then the total number of available characters is:

$$\text{Total Characters} = 26 \text{ (letters)} + 10 \text{ (numbers)} = 36$$

If the system generates a trainer code of length L , the total number of possible unique trainer codes can be calculated as:

$$\text{Total Possible Codes} = 36^L$$

For example, if the trainer code length is 6 characters, then:

<i>Total Possible Codes</i>	<i>Number of possible combinations</i>	<i>Number of possible codes</i>
$= 36^6 = 2,176,782,336$		

This large number of combinations significantly reduces the probability of code duplication and ensures scalability as the number of trainers increases.

The validation process uses database indexing techniques which allow trainer code lookup operations to be performed efficiently with constant time complexity:

$$\text{Time Complexity} = O(1)$$

For example, if a trainer receives the code **TRN4582**, trainees entering this code will automatically be linked to that specific trainer and gain access to assigned workout and diet plans.

5 Gantt chart

Table 1: Gantt chart for La Traino App

Process	No. of Weeks									
	1	2	3	4	5	6	7	8	9	10
Requirement Gathering										
Planning										
Designing										
Coding										
Testing and debugging										
Implementation										

6 Expected outcome

- Functional and interactive physical training platform
- Personalized training experience for trainees
- Easy management of trainees for trainers
- Reduction in manual workout assigning
- Easy progress tracking and feedback from trainer

7 References

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